THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

9 May 2016 TIME: 9:30 – 12:30 (3 hours)

Answer **5 (FIVE)** questions.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Do not use pencil!

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A

1. A simply supported beam AB consists of three portions, i.e. AD and EB that are horizontal and DE that is inclined, as shown in Figure 1. A point load is applied each at mid-span of AD (P_1 = 50kN) and EB (P_2 = 80kN), while a uniformly distributed load (q = 40kN/m) is applied along DE and perpendicular to DE.
   - Determine the reactions at A and B.
   - Draw to scale the axial force diagram (AFD) for beam AB.
   - Draw (on tension side) to scale the bending moment diagram (BMD) for beam AB.
   - Determine the value and location of maximum moment within DE.
   - Draw to scale the shear force diagram (SFD) for beam AB.

   Draw the AFD, BMD and SFD following the shape of the beam.
   (20 marks)

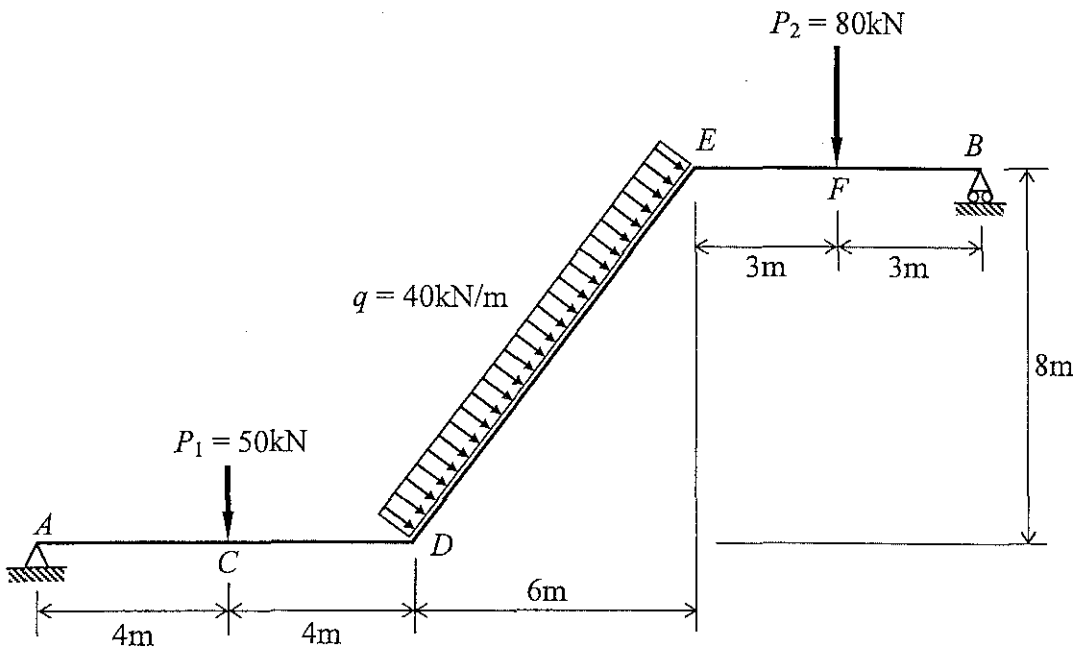
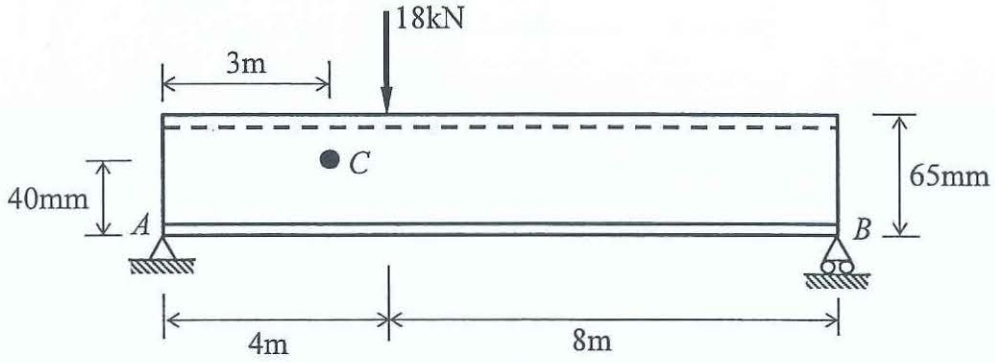


Figure 1

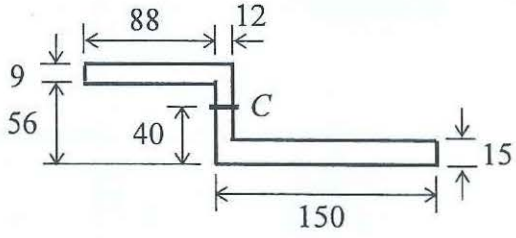
2. A simply supported beam AB is subjected to a point load of 18kN at one-third of its span as shown in Figure 2. The beam has a z-shaped cross section (see Figure 2) and span of 12m. Point C is located 3m to the right of A and 40mm above the base of cross section (see Figure 2).

   For point C: - Determine the directions and values of principal stresses.
   - Determine the direction and value of maximum shear stress.

   (20 marks)



Beam
Not to scale.



Cross section
Not to scale.
All dimensions in mm.

Figure 2

3. (a) Two types of circular solid metal bars are joined concentrically as shown in Figure 3a. The first metal has cross-section diameter d_1 = 25mm, length ℓ_1 = 250mm and Young's modulus E_1 = 180GPa, while the second metal d_2 = 16mm, ℓ_2 = 100mm and E_2 = 80GPa. The two bars are fitted between two fixed supports.
   Determine the stress in each of the bars if the distance between the fixed supports is reduced by 0.4mm. **(8 marks)**

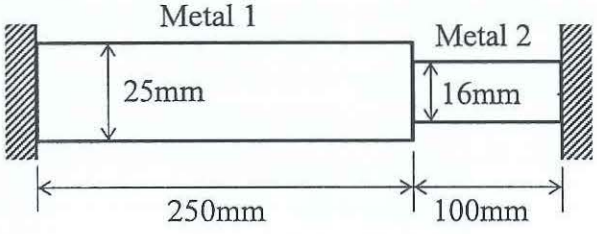


Figure 3a

3. (b) Sketch the bending moment diagram (BMD) and shear force diagram (SFD) for the simply supported beam ABC shown in Figure 3b. The beam is subjected to a clockwise moment at B, where $a < b$. Sketch clearly within the same diagram whether the values are larger, smaller or identical. The BMD must be sketched on tension side. **(8 marks)**.

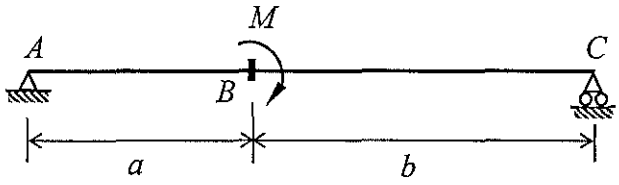


Figure 3b

3. (c) A column AB is subjected to an axial compression force as illustrated in Figure 3c. The column is unrestrained for the whole length, except at its both supports. Predict about which axis the buckling will happen if the cross section of the column is made up of a rectangular section (first cross section in Figure 3c), a T-section (second cross section in Figure 3c), an I-section (third cross section in Figure 3c) or an equal-angle section (fourth cross section in Figure 3c). Draw this axis on each of the cross section. **(4 marks)**

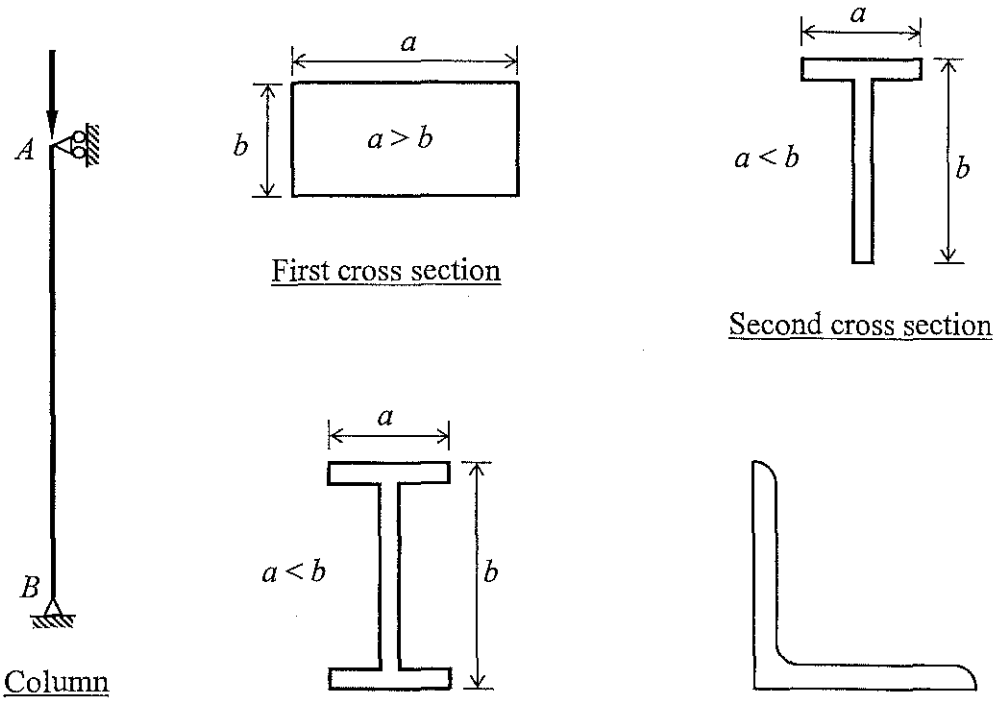


Figure 3c